CIE AS PHYSICS (9702) • QUESTION PAPER

9.2 Potential Difference & Power

Syllabus Section: 9. Electricity **Paper Type:** Multiple Choice (Paper 1) **Total:** 18 Questions (18 Marks)

Question 1 Easy [1 mark]

What describes the electric potential difference between two points in a wire that carries a current?

A the force required to move a unit positive charge between the points

B the ratio of the energy dissipated between the points to the current

C the ratio of the power dissipated between the points to the current

D the ratio of the power dissipated between the points to the charge moved

Question 2 Easy [1 mark]

A filament lamp has a resistance of 180 Ω when the current in it is 500 mA.

What is the power transformed in the lamp?

A 45 W

B 50 W

C 90 W

D 1400 W

Question 3 Easy [1 mark]

A 12 V battery is charged for 20 minutes by connecting it to a source of electromotive force (e.m.f.). The battery is supplied with 7.2×10^4 J of energy in this time.

How much charge flows into the battery?

A 5.0 C

B 60 C

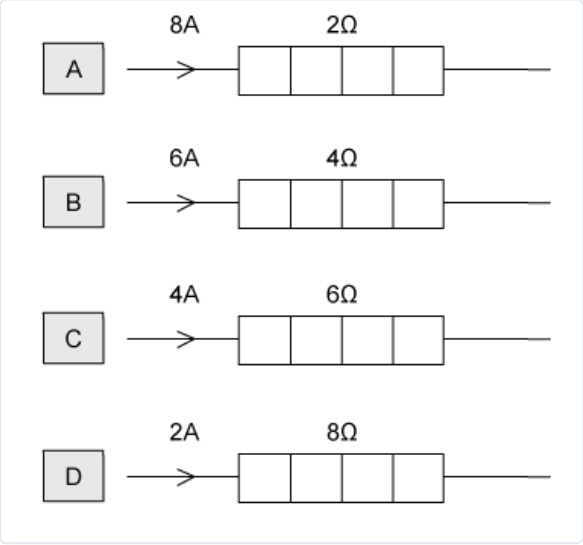
C 100 C

D 6000 C

Question 4 Easy [1 mark]

The diagram shows four heaters and the current in each.

Which heater has the greatest power dissipation ?



A

B

C

D

Question 5 Easy [1 mark]

What is the correct definition for potential difference?

A Energy transferred per unit charge

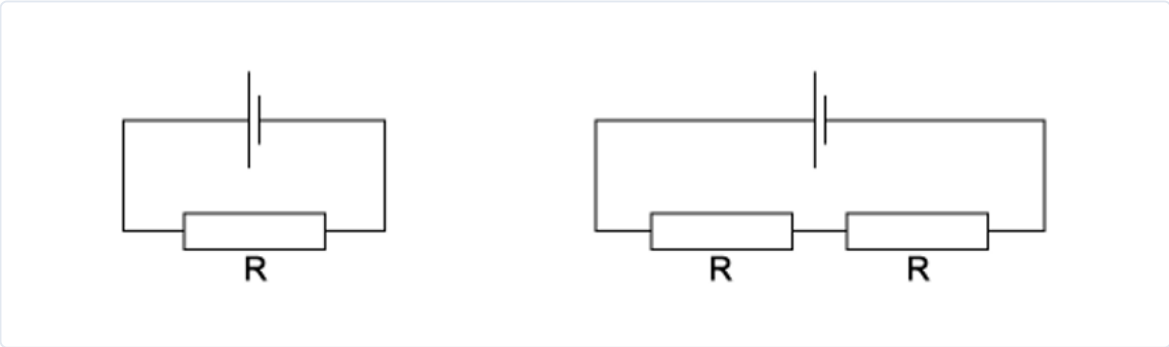
B Energy transferred per unit current

C Power transfer per unit charge

D Work done per unit current

Question 6 Medium [1 mark]

The diagrams show two different circuits.



The cells in each circuit have the same electromotive force and zero internal resistance. The three resistors each have the same resistance *R*.

In the circuit on the left, the power dissipated in the resistor is *P*.

What is the total power dissipated in the circuit on the right?

A $\frac{P}{4}$

B $\frac{P}{2}$

C P

D 2P

Question 7 Medium [1 mark]

The potential difference across a component in a circuit is 2.0 V.

How many electrons must flow through this component in order for it to be supplied with 4.8 J of energy?

A 2.6×10^{18}

B 1.5×10^{19}

C 3.0×10^{19}

D 6.0×10^{19}

Question 8 Medium [1 mark]

Two lamps are connected in series to a 250V power supply. One lamp is rated 240V, 60W and the other is rated 10V, 2.5W.

Which statement most accurately describes what happens ?

A both lamps light at less than their normal brightness

B both lamps light normally

C only the 60 W lamp lights

D the 10 V lamp blows

Question 9 Medium [1 mark]

A power cable X has resistance *R* and carries current I.

A second cable Y has resistance 2*R* and carries current $\frac{I}{2}$

What is the ratio $\frac{power\ \ dissipated\ \ in\ \ Y}{power\ \ dissipated\ \ in\ \ X}$?

A $\frac{1}{4}$

B $\frac{1}{2}$

C 2

D 4

Question 10 Medium [1 mark]

In a simple electrical circuit, the current in a resistor is measured as (2.50 ± 0.05) mA. The resistor is marked as having a value of 4.7Ω ± 2 %.

If these values were used to calculate the power dissipated in the resistor, what would be the percentage uncertainty in the value obtained?

A 2 %

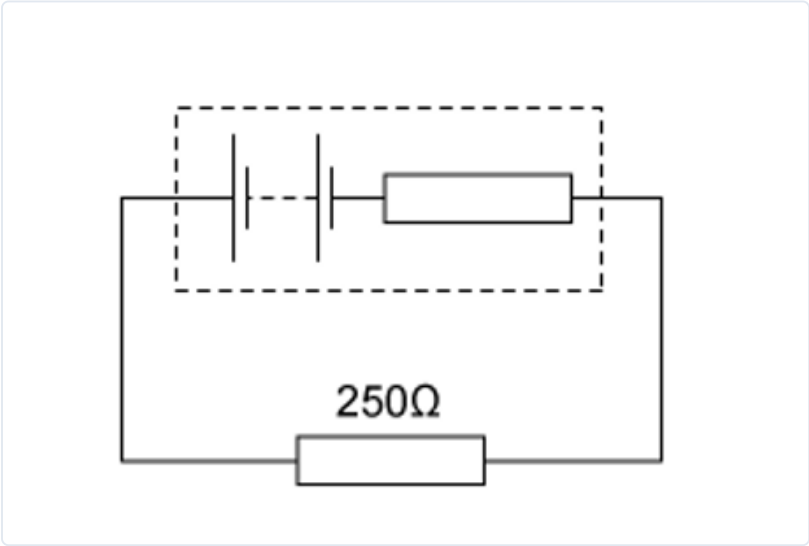
B 4 %

C 6 %

D 8 %

Question 11 Medium [1 mark]

A battery, with a constant internal resistance, is connected to a resistor of resistance 250 Ω, as shown.



The current in the resistor is 40 mA for a time of 60 s. During this time 6.0 J of energy is lost in the internal resistance.

What is the energy supplied to the external resistor during the 60 s and the e.m.f of the battery ?

	Energy / J	E. m. f / V
A	2.4	2.4
B	2.4	7.5
C	24	10.0
D	24	12.5

A

B

C

D

Question 12 Medium [1 mark]

The charge that an electric battery can deliver is specified in ampere-hours.

For example, a battery of capacity 40 ampere-hours could supply, when fully charged, 0.2 A for 200 hours.

What is the maximum energy that a fully charged 12 V, 40 ampere-hour battery could supply ?

A 1.7 kJ

B 29 kJ

C 1.7 MJ

D 29 MJ

Question 13 Hard [1 mark]

The amount of energy transferred when 10 C of charge passes through a p. d. of 20 V is the same as the energy needed to raise a 2 kg mass through a distance *x*.

Take gravitational field strength to be 10 N kg^{-1}.

What is the value of *x*?

A 0.1 m

B 1.0 m

C 10 m

D 100 m

Question 14 Hard [1 mark]

The ratings of two lamps are as follows:

X	230 V, 100 W
Y	230 V, 60 W

During normal operation, which of the lamps will have larger current and resistance?

A $I_X > I_Y \quad and \quad R_X > R_Y$

B $I_X > I_Y \quad and \quad R_X < R_Y$

C $I_X < I_Y \quad and \quad R_X > R_Y$

D $I_X < I_Y \quad and \quad R_X < R_Y$

Question 15 Hard [1 mark]

Two lightbulbs, **X** and **Y**, are in parallel and are attached to a cell which has a potential difference of 75 V. Lightbulb **X** has a resistance of 350 Ω and Lightbulb **Y** has a resistance of 500 Ω. The power of Lightbulbs **X** and **Y** are P_X and P_Y respectively.

What is the value of the ratio $\frac{P_X}{P_Y}$?

A $\frac{1}{2}$

B $\frac{7}{10}$

C $\frac{10}{7}$

D 2

Question 16 Hard [1 mark]

What is the effect on the current and power of a fixed resistor when the voltage across the resistor doubles?

	Current	Power
A.	Quadruples	Doubles
B.	Doubles	Doubles
C.	Doubles	Quadruples
D.	Halves	Doubles

A

B

C

D

Question 17 Hard [1 mark]

Two resistors, **X** and **Y**, are in series with a cell that has a potential difference of *V*. The resistance of **X** and **Y** is *R* and $\frac{R}{2}$ respectively.

Given that the charge through both resistors is 6 C, what is the power and work done of Resistor **X**?

	Power	Work done
A.	2V	$\frac{8V^2}{9R}$
B.	3V	$\frac{V^2}{2R}$
C.	4V	$\frac{4V^2}{9R}$
D.	6V	$\frac{V^2}{R}$

A

B

C

D

Question 18 Hard [1 mark]

A battery with an electromotive force (e.m.f.) of 240 V and an internal resistance of 2.0 Ω is connected to a heating element. The element draws a current of 8.7 A.

The power dissipated by the heating element is the same as the power that would be dissipated by a 12 Ω resistor carrying a current of *I*.

What is the value of *I*?

A 1.3 A

B 4.1 A

C 13 A

D 130 A